



AIRCRAFT MAINTENANCE MANUAL

AIR DISTRIBUTION, CABIN – ADJUSTMENT/TEST

**ON A/C ALL

TASK 21-22-00-710-801

Operational Test of the Passenger Service Unit (PSU) Panel Gasper

1. General

- A. The maintenance procedure that follows is for the operational test of the gaspers at each passenger service unit (PSU) panel. The gaspers are installed in the PSU panels above each passenger seat in the passenger compartment.

2. Job Set-Up Information

Subtask 21-22-00-946-001

A. Reference Information

REFERENCE	DESIGNATION
TASK 24-00-00-861-801	Energize the Electrical System
TASK 24-00-00-861-802	De-energize the Electrical System
TASK 24-00-00-910-801	Electrical/Electronic Safety Precautions

3. Job Set-Up

Subtask 21-22-00-910-001

WARNING: OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

- A. Obey all the electrical/electronic safety precautions (Refer to [TASK 24-00-00-910-801](#)).

Subtask 21-22-00-865-001

- B. Make sure that the circuit breakers that follow are closed:

CB PANEL AND CB NO	NAME
RIGHT DC (MAIN), Q6	ECS/BA CONT – L MAN
RIGHT DC (MAIN), R6	ECS/BA CONT – R AUTO
LEFT DC (MAIN), B7	ECS/BA CONT – L AUTO
LEFT DC (MAIN), C7	ECS/BA CONT – R MAN



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Subtask 21-22-00-861-001

- C. Energize the aircraft electrical system (Refer to [TASK 24-00-00-861-801](#)).

4. Procedure

Subtask 21-22-00-710-001

- A. Do an operational test of the gasper at each PSU panel as follows:
- (1) Make sure that the BLEED 1 and BLEED 2 switches on the air conditioning control panel are in the OFF position.
 - (2) Open all the gaspers on the PSU panels.
 - (3) Set the RECIRC fan switch on the air conditioning control panel to the RECIRC position to energize the recirculation fan.
 - (4) Make sure that the recirculation fan operates.
 - (5) Do a check of the airflow from each gasper. Make sure that you feel a sufficient air flow from each gasper.
 - (6) If there is not a sufficient airflow from a gasper, examine the gasper feed lines for a kink or blockage.
 - (7) Set the RECIRC fan switch on the air conditioning control panel to the off position.

5. Close-Out

Subtask 21-22-00-861-002

- A. De-energize the aircraft electrical system (Refer to [TASK 24-00-00-861-802](#)).

Subtask 21-22-00-941-001

- B. Remove all tools, equipment, and unwanted materials from the work area.



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FAN, RECIRCULATION – ADJUSTMENT/TEST

**ON A/C ALL

TASK 21-22-16-710-801

Operational Test of the Recirculation Fan

1. General

- A. The maintenance procedures that follow are for the operational test of the recirculation fan. There is one recirculation fan located at the bottom of the air conditioning pack in the aft fuselage.

2. Job Set-Up

Subtask 21-22-16-865-002

- A. Make sure the circuit breakers that follow are closed:

CB PANEL AND CB NO.	NAME
RIGHT DC, Q6	ECS/BA – L MAN
RIGHT DC, R6	ECS/BA – R AUTO
LEFT DC, B7	ECS/BA – L AUTO
LEFT DC, C7	ECS/BA – R MAN

3. Procedure

Subtask 21-22-16-710-002

- A. Do an operational test of the recirculation fan as follows:
- (1) On the AIR CONDITIONING panel, set the controls to the positions that follows:
 - (a) Select the RECIRC switch to RECIRC position.
 - (b) PACK switches may remain in AUTO position.
 - (c) Allow the system to stabilize for five minutes minimum.
 - (2) Make sure that there is no air leakage from the fan assembly.
 - (3) Check for the environmental control unit (ECU) fault codes with the centralized diagnostics system (CDS) as follows:
 - (a) The CDS maintenance panel is installed above the forward wardrobe. Remove the wardrobe cover for access.
 - (b) To operate the CDS in the maintenance mode make sure that the indicated airspeed is less than 40 kts and the aircraft is in a weight-on-wheels condition.
 - (c) Set the CDS GND MAINT switch to the ON position to energize the panel and make sure that the amber light comes on.



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- (d) If necessary, clear the old fault messages from the ECS as follow:
1. On the CDS GND MAINT panel set the rotary selector knob to the ECS position.
 2. On the CDS GND MAINT panel push the NVM RESET switch for a minimum of two seconds to erase the fault codes from the ECU memory.
 3. Make sure that the ECS AUTO FAIL light is off. The light comes on when a failure is recorded in the ECS system.

NOTE: This indication is only shown with a weight-on-wheels (WOW) condition.
 4. Set the rotary selector knob to the OFF position.
- (e) In the flight-compartment aft center-console, set the left audio radio control display unit (ARCDU) selector switch to the ON position.
- (f) Press the MAINT (maintenance) key on the left ARCDU. This will display the CDS main page menu items; AVIONICS, OTHER SYSTEMS and AVIONIC DATA.
- (g) On the ARCDU panel, push the 2L key to get the OTHER SYSTEMS menu.
- (h) Push the 3L key adjacent to the ECS message to show the ECS PRESENT FLT L CH 1/7 page for the left channel fault data.
- (i) Push the NEXT key to show the ECS PRESENT FLT R CH 2/7 page for the right channel fault data. Push the PREV key to go back to the previous menu page.
- (j) Make sure that the fault codes listed below for the recirculation fan are not shown:
- 3C01-RECIRC FAN
 - 3C02-RECIRC FAN
- (k) For all other fault messages refer to the fault isolation manual.
- (l) On the ARCDU push the DIM/RTN key to return to the main page.
- (m) In the flight-compartment aft center-console, set the audio radio control display unit (ARCDU) selector switch to the OFF position.
- (n) Put the CDS GND MAINT switch to the OFF position. Make sure that the amber light is off.
- (4) On the AIR CONDITIONING panel, set the controls to the positions that follows:
- (a) Set the left and right PACKS switches to the OFF position.
 - (b) Select the RECIRC switch to the OFF position.



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4. Close Out

Subtask 21-22-16-941-002

- A. Remove all tools, equipment and unwanted materials from the work area.

PSM 1-84-2
EFFECTIVITY:
See first effectivity on page 501 of 21-22-16

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TEMPERATURE SWITCHES, ZONE – ADJUSTMENT/TEST

**ON A/C ALL

TASK 21-25-09-720-801

Functional Test of the Zone Temperature Switch

1. General

- A. The maintenance procedure that follows is for the functional test of the zone temperature switch (written as the "temperature switch" in this procedure). The temperature switch is installed in the flight compartment forward of the console panel. There is one temperature switch on the pilot side and one on the copilot side.

2. Job Set-Up Information

Subtask 21-25-09-943-001

A. Tools & Equipment

- (1) Commercially Available Heat Gun

Subtask 21-25-09-944-001

B. Consumable Materials

- (1) 13-04 Mist, Freeze

Subtask 21-25-09-946-003

C. Reference Information

REFERENCE	DESIGNATION
TASK 24-00-00-861-801	Energize the Electrical System
TASK 24-00-00-861-802	De-energize the Electrical System
TASK 24-00-00-910-801	Electrical/Electronic Safety Precautions
TASK 32-00-00-840-804	Standard Configuration for an 'On-Ground' Weight-Off-Wheels (WOW) Condition

3. Job Set-Up

Subtask 21-25-09-910-002

- A. Obey all the electrical/electronic safety precautions (Refer to [TASK 24-00-00-910-801](#)).



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Subtask 21-25-09-861-002

- B. Energize the aircraft electrical system (Refer to [TASK 24-00-00-861-801](#)).

Subtask 21-25-09-865-003

- C. Make sure that the circuit breakers that follow are closed:

CB PANEL AND CB NO	NAME
LEFT (ESSENTIAL), M6	FAN 1 PWR SEC
LEFT (MAIN), A6	FAN 1 PWR PRI
LEFT (MAIN), B6	FAN 3 PWR SEC
RIGHT (ESSENTIAL), K9	COOLING FAN CONT
RIGHT (ESSENTIAL), L9	FAN 2 PWR SEC
RIGHT (MAIN), R4	FAN 2 PWR PRI
RIGHT (MAIN), S4	FAN 3 PWR PRI

4. Procedure

Subtask 21-25-09-720-002

- A. Do the functional test of the temperature switch as follows:
- (1) Put the aircraft in the on-ground in the weight-off-wheels condition (Refer to [TASK 32-00-00-840-804](#)).
 - (2) Make sure that the fan 1 and fan 2 operate in the low speed mode.
 - (3) Make sure that the fan 3 is off.
 - (4) Disconnect the electrical connectors 2126-P5 and 2126-P6 from the left and the right temperature switches.
 - (5) Make sure that the fan 1 and fan 2 operate in the high speed mode.
 - (6) Make sure that the fan 3 is off.
 - (7) Ground the pin 2 of the left and the right temperature switches.
 - (8) Put the aircraft in the weight-on-wheels condition (Refer to [TASK 32-00-00-840-804](#)).
 - (9) Make sure that the fan 1, fan 2 and fan 3 are off.
 - (10) Remove the ground from the pin 2 of the right temperature switch.
 - (11) Make sure that the fan 1 is off.
 - (12) Make sure that the fan 2 and fan 3 operate in the high speed mode.



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- (13) Ground the pin 2 of the right temperature switch.
- (14) Remove the ground from the pin 2 of the left temperature switch.
- (15) Make sure that the fan 1 operates in the high speed mode.
- (16) Make sure that the fan 2 and fan 3 are off.
- (17) Remove the ground from the right temperature switch.
- (18) Connect the electrical connectors 2126-P5 and 2126-P6 to the left and right temperature switches.
- (19) Make sure that the fan 1 and fan 2 operate in the low speed mode.
- (20) Make sure that the fan 3 is off.
- (21) Carefully heat the left temperature switch sensor with a heat gun.
NOTE: The sensor is the pin installed on the opposite side of the temperature switch connector. Do not heat the connector or the main body of the temperature switch.
- (22) Make sure that the fan 1 operates in the high speed mode.
- (23) Make sure that the fan 2 operates in the low speed mode.
- (24) Make sure that the fan 3 is off.
- (25) Remove the heat gun from the left temperature switch sensor.
- (26) Cool the left temperature switch sensor using freeze mist until the fan 1 operates in the low speed mode.
NOTE: Do not freeze the temperature switch sensor.
- (27) Make sure that the fan 1 and fan 2 operate in the low speed mode.
- (28) Make sure that the fan 3 is off.
- (29) Carefully heat the right temperature switch sensor with a heat gun.
NOTE: The sensor is the pin installed on the opposite side of the temperature switch connector. Do not heat the connector or the main body of the temperature switch.
- (30) Make sure that the fan 1 operates in the low speed mode.
- (31) Make sure that the fan 2 operates in the high speed mode.
- (32) Make sure that the fan 3 is off.
- (33) Remove the heat gun from the right temperature switch sensor.
- (34) Cool the right temperature switch sensor using freeze mist until the fan 2 operates in the low speed mode.
NOTE: Do not freeze the temperature switch sensor.
- (35) Make sure that the fan 1 and fan 2 operate in the low speed mode.
- (36) Make sure that the fan 3 is off.



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5. Close Out

Subtask 21-25-09-861-001

- A. De-energize the aircraft electrical system (Refer to [TASK 24-00-00-861-802](#)).

Subtask 21-25-09-941-002

- B. Remove all the tools, equipment, and unwanted materials from the work area.



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**ON A/C ALL

TASK 21-31-00-720-801

Functional Test of the Pressurization Control

1. General

- A. The maintenance procedure that follows is for the functional test of the pressurization control system.

2. Job Set-Up Information

Subtask 21-31-00-946-002

A. Reference Information

REFERENCE	DESIGNATION
TASK 24-00-00-861-801	Energize the Electrical System
TASK 24-00-00-861-802	De-energize the Electrical System
TASK 24-00-00-910-801	Electrical/Electronic Safety Precautions
TASK 49-00-00-869-801	Start Procedure of the APU
TASK 49-00-00-869-802	Shut Down Procedure of the APU
TASK 71-00-00-868-801	Engine Start
TASK 71-00-00-868-802	Engine Shutdown

3. Job Set-Up

Subtask 21-31-00-910-002

WARNING: OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

- A. Obey all the electrical/electronic safety precautions (Refer to [TASK 24-00-00-910-801](#)).

Subtask 21-31-00-861-015

- B. Energize the electrical system (Refer to [TASK 24-00-00-861-801](#)).

Subtask 21-31-00-010-017

- C. For access to the outflow valve, open the door that follows:



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REFERENCE	DESIGNATION
311AB	Door

4. Procedure

Subtask 21-31-00-720-002

- A. Do the functional test of the pressurization control system in manual mode as follows:
- (1) On the cabin pressurization control panel (CPCP), set the AUTO/MAN/DUMP switch to the MAN position.
 - (2) On the CPCP, set the MAN DIFF switch to the INCR position for 60 seconds.
 - (a) Make sure the outflow valve goes to the full closed position.
 - (3) On the CPCP, set the MAN DIFF switch to the DECR position for 60 seconds.
 - (a) Make sure the outflow valve goes to the full open position.
 - (4) On the CPCP, set the MAN DIFF switch to the INCR position for 60 seconds.
 - (a) Make sure the outflow valve goes to the full closed position.
 - (5) On the CPCP, set the AUTO/MAN/DUMP switch to the DUMP position.
 - (a) Make sure the outflow valve goes to the full open position.

Subtask 21-31-00-720-023

- B. Do the functional test of the pressurization control system in the automatic mode as follows:
- (1) Make sure all the doors are closed and locked.
 - (2) On the CPCP, set the AUTO/MAN/DUMP switch to the AUTO position.
 - (3) Set the LDG ALT knob to get a LDG ALT of 0 ft.
 - (4) Start the engines (Refer to [TASK 71-00-00-868-801](#)) or the APU (Refer to [TASK 49-00-00-869-801](#)) to supply a source of bleed air.
 - (5) Set the throttle levers to the IDLE position.
 - (6) If the APU is used as the source of bleed air, do as follows:
 - (a) On the AIR CONDITIONING panel, make sure that:
 - The BLEED 1 switch is set to the OFF position
 - The BLEED 2 switch is set to the OFF position.
 - (b) On the APU CONTROL panel, push the APU BL AIR switch. Make sure that the BL AIR OPEN light comes on.
 - (7) If the engines 1 and 2 are used as the source of bleed air, do as follows:



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- (a) On the AIR CONDITIONING panel, set the switches that follow:
 - The BLEED 1 switch to the 1 position
 - The BLEED 2 switch to the 2 position.
- (8) On the AIR CONDITIONING panel, set the switches that follow:
 - The BLEED knob to the NORM position
 - The PACKS switches to the AUTO position.
- (a) Through door 311AB, make sure that the aft safety valve goes to the open position.
- (9) Set the engine throttle levers above the 60° position (minimum takeoff power).
 - (a) On the overhead panel, make sure that the RATE is at –300 ft/min until the ALT is at –400 feet from the ambient.
- (10) Set the engine throttle levers to the IDLE position.
 - (a) On the overhead panel, make sure that the RATE indication is at +500 ft/min until the ALT is at the ambient.
- (11) On the AIR CONDITIONING panel, set the PACKS switches to the OFF position.
- (12) If the APU is on, push the APU BL AIR switch on the APU CONTROL panel. Make sure that BL AIR OPEN light goes off.
- (13) Shut down the engines (Refer to [TASK 71-00-00-868-802](#)) or the APU (Refer to [TASK 49-00-00-869-802](#)).

5. Close Out

Subtask 21-31-00-941-030

- A. Remove all tools, equipment, and unwanted materials from the work area.

Subtask 21-31-00-410-017

- B. Close the door that follows:

REFERENCE	DESIGNATION
311AB	Door

Subtask 21-31-00-861-017

- C. De-energize the electrical system (Refer to [TASK 24-00-00-861-802](#)).



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VALVE, OUTFLOW – ADJUSTMENT/TEST

**ON A/C ALL

TASK 21-31-16-720-801

Functional Test of the Aft Outflow Valve

1. General

A. The maintenance procedure that follows is for the functional test of the aft outflow valve.

2. Job Set-Up Information

Subtask 21-31-16-946-003

A. Reference Information

REFERENCE	DESIGNATION
TASK 12-00-06-861-801	Connection of the External DC Electrical Power
TASK 12-00-06-861-803	Connection of the External AC Electrical Power
TASK 12-00-06-861-802	Removal of the External DC Electrical Power
TASK 12-00-06-861-804	Removal of the External AC Electrical Power
TASK 24-00-00-910-801	Electrical/Electronic Safety Precautions

3. Job Set-Up

Subtask 21-31-16-910-001

A. Obey all the electrical/electronic safety precautions (Refer to [TASK 24-00-00-910-801](#)).

Subtask 21-31-16-861-001

B. Connect and energize the external DC power (Refer to [TASK 12-00-06-861-801](#)).

Subtask 21-31-16-861-002

C. Connect and energize the external AC power (Refer to [TASK 12-00-06-861-803](#)).

Subtask 21-31-16-865-003

D. Make sure that the circuit breaker that follows is closed:

CB PANEL AND CB NO.	NAME
LEFT DC (ESSENTIAL), L5	CABIN PRESS MAN



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4. Procedure

Subtask 21-31-16-720-002

A. Do the functional test of the aft outflow valve as follows:

- (1) On the cabin pressurization control panel, set the AUTO/MAN/DUMP switch to the MAN position.
- (2) On the cabin pressurization control panel, set the MAN DIFF switch to INCR and hold for 60 seconds.
 - (a) Visually make sure that the outflow valve goes to the FULL CLOSED position.
- (3) On the cabin pressurization control panel, set the MAN DIFF switch to DECR and hold for 60 seconds.
 - (a) Visually make sure that the outflow valve goes to the FULL OPEN position.
- (4) On the cabin pressurization control panel, set the MAN DIFF switch to INCR and hold for 60 seconds.
 - (a) Visually make sure that the outflow valve goes to the FULL CLOSED position.
- (5) On the cabin pressurization control panel, set the AUTO/MAN/DUMP switch to DUMP.
 - (a) Visually make sure that the outflow valve goes to the FULL OPEN position.

5. Close Out

Subtask 21-31-16-861-003

A. Remove the external DC power (Refer to [TASK 12-00-06-861-802](#)).

Subtask 21-31-16-861-004

B. Remove the external AC power (Refer to [TASK 12-00-06-861-804](#)).

Subtask 21-31-16-941-002

C. Remove all tools, equipment, and unwanted materials from the work area.



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AIR CONDITIONING PACK – SERVICING

**ON A/C WITH HEAT EXCHANGERS P/N 820905-5 ONLY

TASK 21-51-00-900-801

Restoration of the Heat Exchanger by Cleaning and Functional Check (Primary and Secondary Circuit Leakage Tests) (MRB#215100-208)

1. General

- A. The maintenance procedure that follows is for the restoration of the Environmental Control System (ECS) dual heat-exchanger.
- B. This procedure is applicable to heat exchangers with part number 820905-5 only.

2. Job Set-Up Information

Subtask 21-51-00-946-008

A. Reference Information

REFERENCE	DESIGNATION
TASK 21-51-01-000-801	Removal of the Dual Heat Exchanger
TASK 21-51-01-400-801	Installation of the Dual Heat Exchanger

3. Procedure

Subtask 21-51-00-020-007

- A. Remove the dual heat exchanger (Refer to [TASK 21-51-01-000-801](#)).

Subtask 21-51-00-300-001

- B. Send the heat exchanger to the manufacturer (or an approved repair facility) for restoration (shop cleaning and functional checks).

Subtask 21-51-00-420-007

- C. Install a new or serviceable dual heat-exchanger (Refer to [TASK 21-51-01-400-801](#)).

4. Close Out

Subtask 21-51-00-941-008

- A. Remove all tools, equipment, and unwanted materials from the work area.

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**ON A/C ALL

TASK 21-61-00-710-803

Operational Test of the ECS Temperature Control

1. General

- A. The maintenance procedures that follow are for an operational test of the environmental control system (ECS) temperature control functions.

2. Job Set-Up Information

Subtask 21-61-00-946-002

A. Reference Information

REFERENCE	DESIGNATION
TASK 12-00-06-861-801	Connection of the External DC Electrical Power
TASK 12-00-06-861-802	Removal of the External DC Electrical Power
TASK 24-00-00-910-801	Electrical/Electronic Safety Precautions
TASK 49-00-00-869-801	Start Procedure of the APU
TASK 49-00-00-869-802	Shut Down Procedure of the APU

3. Job Set-Up

Subtask 21-61-00-910-003

- A. Obey the electrical/electronic safety precautions (Refer to [TASK 24-00-00-910-801](#)).

Subtask 21-61-00-865-006

- B. Make sure that the circuit breakers that follow are closed:

CB PANEL AND CB NO	NAME
LEFT DC, B7	ECS/BA CONT L AUTO
LEFT DC, C7	ECS/BA CONT R MAN
RIGHT DC, R6	ECS/BA CONT R AUTO
RIGHT DC, Q6	ECS/BA CONT L MAN

Subtask 21-61-00-861-003

- C. Energize the DC electrical system (Refer to [TASK 12-00-06-861-801](#)).



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4. Procedure

Refer to Figure 503.

Subtask 21-61-00-710-007

- A. Do an operational test of the ECS temperature control system as follows:
- (1) On the cabin pressure control panel, set the AUTO/MAN/DUMP switch to the DUMP position. This will open the aft outflow valve to prevent aircraft pressurization. If necessary, open a cabin door during the operational test.
 - (2) On the ICE PROTECTION panel, set the AIRFRAME MODE SELECT switch to the OFF position.
 - (3) Start the APU (Refer to [TASK 49-00-00-869-801](#)).
 - (4) On the APU CONTROL panel, press the BL AIR switch and make sure the BL AIR OPEN light comes on.
 - (5) On the AIR CONDITIONING panel, make sure the switches are in the positions shown:
 - The 1 BLEED switch (4) in the OFF position
 - The 2 BLEED switch (4) in the OFF position
 - The left PACKS switch (7) in the AUTO position
 - The right PACKS switch (7) in the AUTO position
 - The CABIN TEMP CONTROL knob (3) in the mid range position
 - The FLT COMP TEMP CONTROL knob (8) in the mid range position
 - The GAUGE knob (6) to the CABIN position
 - The BLEED switch (5) to the NORM position
 - The RECIRC switch (1) in the ON position.
 - (6) Allow the ECS system to become stable for 5 minutes.
 - (7) On the AIR CONDITIONING panel, with the TEMP CONTROL knobs in the mid-range position, record the CAB DUCT and FC DUCT temperatures on the DUCT TEMP gauge.
 - (8) Turn the CABIN and FLT COMP TEMP CONTROL knobs (3) and (8) to the full COOL position.
 - (9) Wait approximately 2 to 3 minutes, then with the GAUGE knob (6) check and record the temperatures for the CAB DUCT and FC DUCT positions shown on the duct temperature gauge (2).
 - (10) Make sure that the temperature indicated is less than the temperature recorded at the mid-range position.
 - (11) Turn the CABIN and FLT COMP TEMP CONTROL knobs (3) and (8) to the full WARM position.



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- (12) Wait approximately 2 to 3 minutes, then with the GAUGE knob (6) check and record the temperatures for the CAB DUCT and FC DUCT positions.
- (13) Make sure that the temperature indicated has increased.
- (14) Turn the FLT COMP TEMP CONTROL knob (8) to the mid-range position.
- (15) Turn the CABIN TEMP CONTROL knob (3) to the F/A position.
- (16) On the forward flight-attendant control panel, push the DISPLAY TEST switch (10) and make sure you get the indications that follow:
 - The MID CABIN TEMP indicator (9) comes on
 - The WARM /COOL indicator (19) light comes on.
- (17) On the forward flight-attendant control panel, push the TEMP DISPLAY switch (11).
- (18) Push the UP arrow switch (12) until the WARM/COOL indicator (14) goes to full WARM.
- (19) After a few minutes, make sure you get the conditions and indications that follow:
 - (a) In the cabin, warm air flows out of the upper and lower riser outlets.
 - (b) On the forward flight-attendant control panel, the MID CABIN TEMP indication (9) increases.
- (20) Push the DOWN arrow switch (13) until the WARM/COOL indicator (14) goes to full COOL.
- (21) After a few minutes, make sure you get the conditions and indications that follow:
 - (a) In the cabin, cool air flows out of the upper and lower riser outlets.
 - (b) On the forward flight-attendant control panel, the MID CABIN TEMP indication (9) decreases.
- (22) Push the UP or DOWN arrow switches until the WARM/COOL indicator (14) goes to the middle position.
- (23) On the AIR CONDITIONING panel, set the switches to the positions that follow:
 - The left PACKS switch (7) to the OFF position
 - The right PACKS switch (7) to the OFF position
 - The CABIN TEMP CONTROL knob (3) in the mid range position
 - The FLT COMP TEMP CONTROL knob (8) in the mid range position
 - The RECIRC switch (1) to the OFF position.
- (24) Shut down the APU (Refer to [TASK 49-00-00-869-802](#)).
- (25) On the cabin pressure control panel, set the AUTO/MAN/DUMP switch to the AUTO position.



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5. Close Out

Subtask 21-61-00-861-004

- A. De-energize the DC electrical system (Refer to [TASK 12-00-06-861-802](#)).

Subtask 21-61-00-941-002

- B. Remove all tools, equipment, and unwanted materials from the work area.

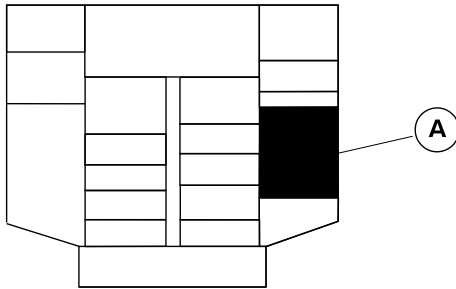
PSM 1-84-2
EFFECTIVITY:
See first effectivity on page 510 of 21-61-00

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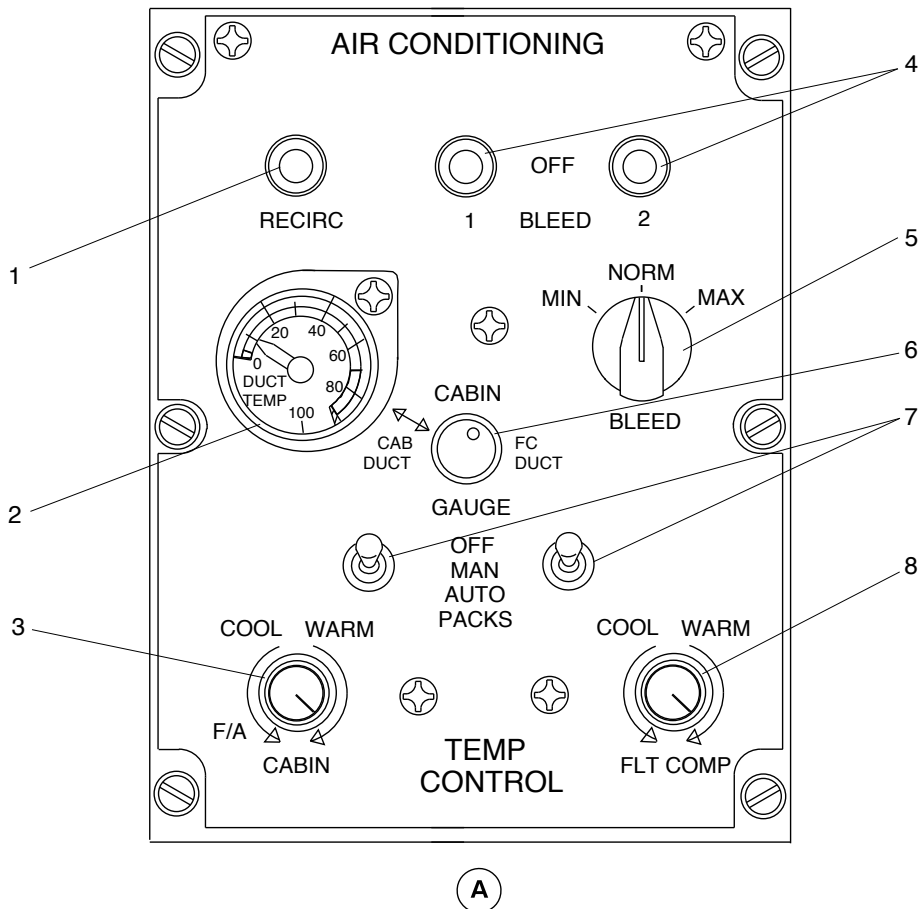
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LEGEND

1. RECIRC switch.
2. DUCT TEMP gauge.
3. CABIN TEMP CONTROL knob.
4. BLEED switches.
5. BLEED switch.
6. GAUGE knob.
7. PACKS switches.
8. FLT COMP TEMP CONTROL knob.



brb57a01.dwg, gw, 10/02/06

Operational Test of the ECS Temperature Control – Adjustment/Test
Figure 503 (Sheet 1 of 2)

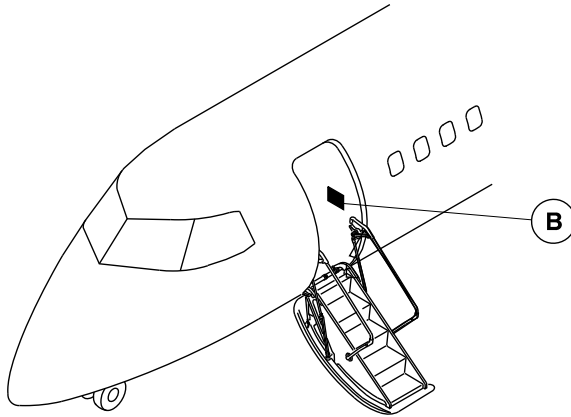
PSM 1-84-2
EFFECTIVITY:
See first effectivity on page 510 of 21-61-00

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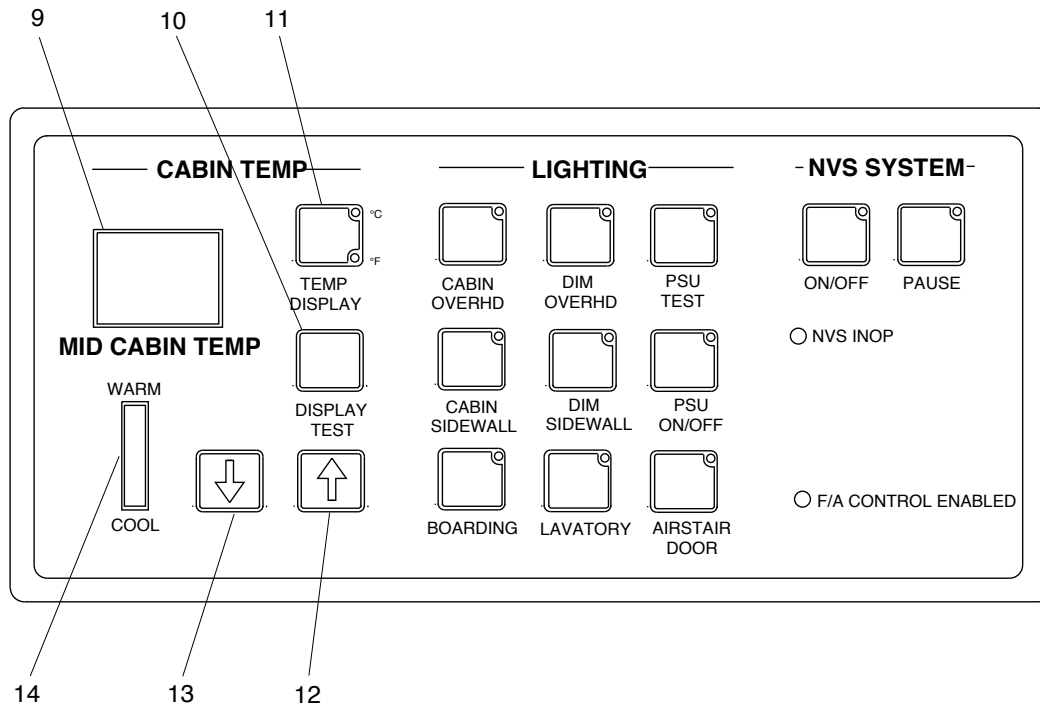
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LEGEND

- 9. Mid cabin temp indicator.
- 10. Display test switch.
- 11. Temp display switch.
- 12. UP arrow switch.
- 13. DOWN arrow switch.
- 14. Warm/Cool indicator.



(B)

brbb57 a02.dg, gw, 10/02/06

Operational Test of the ECS Temperature Control – Adjustment/Test
Figure 503 (Sheet 2 of 2)

PSM 1-84-2
EFFECTIVITY:
See first effectivity on page 510 of 21-61-00

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AIRCRAFT MAINTENANCE MANUAL

TEMPORARY REVISION 21-123

FILING INSTRUCTIONS:

Insert this Temporary Revision after page 501 of AMM TASK 21-00-00-863-801.

Record the insertion of this Temporary Revision on the yellow RECORD OF TEMPORARY REVISIONS SHEET at the front of the Chapter.

Keep this Temporary Revision until normal revision service incorporates the information into the Aircraft Maintenance Manual.

DESCRIPTION:

This Temporary Revision revises AMM TASK 21-00-00-863-801 to revise the procedural steps 4.A.(1).(k).4 and 4.B.(1).(k).4.



AIRCRAFT MAINTENANCE MANUAL

AIR CONDITIONING, GENERAL – ADJUSTMENT/TEST

**ON A/C ALL

TASK 21-00-00-863-801

Pressurize the Aircraft for Maintenance

1. General

- A. The maintenance procedure that follows is for the pressurization of the aircraft for maintenance tasks.

2. Job Set-Up Information

**ON A/C 4001, 4006, 4008-4015, 4018-4037

Subtask 21-00-00-943-001

A. Tools & Equipment

- | | | |
|-----|------------------------|--|
| (1) | Commercially available | Stop watch |
| (2) | Commercially available | Calibrated pressure indicator (0 to 25 psi or 0 to 172.37 kPa), absolute pressure or gauge pressure, hand held |

**ON A/C 4003-4004, 4016-4017, 4038-4140, 4142-4199, 4201-4999

Subtask 21-00-00-943-002

B. Tools & Equipment

- | | | |
|-----|------------------------|------------|
| (1) | Commercially available | Stop watch |
|-----|------------------------|------------|

**ON A/C ALL

Subtask 21-00-00-946-001

C. Reference Information

REFERENCE	DESIGNATION
TASK 12-10-52-863-801	Pressurization of the Reservoir Tank for the Door Seal System
TASK 21-00-00-863-802	Depressurize the Aircraft after Maintenance
TASK 24-00-00-861-801	Energize the Electrical System
TASK 24-00-00-861-802	De-energize the Electrical System
TASK 24-00-00-910-801	Electrical/Electronic Safety Precautions

PSM 1-84-2
EFFECTIVITY: See 21-00-00 Page 2 of 10

21-00-00

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TASK 49-00-00-869-801	Start Procedure of the APU
TASK 71-00-00-868-801	Engine Start

3. Job Set-Up

Subtask 21-00-00-910-001

- A. Obey all the electrical/electronic safety precautions (Refer to [TASK 24-00-00-910-801](#)).

Subtask 21-00-00-861-001

WARNING: MAKE SURE YOU TELL ALL PERSONS IN THE AREA OF THE AIRCRAFT THAT YOU WILL ENERGIZE THE AIRCRAFT ELECTRICAL SYSTEM, BEFORE YOU TURN ON THE ELECTRICAL SYSTEM. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONNEL AND/OR DAMAGE TO EQUIPMENT.

- B. Energize the electrical system (Refer to [TASK 24-00-00-861-801](#)).

Subtask 21-00-00-865-001

- C. Make sure the following circuit breakers are closed:

CB PANEL AND CB NO	NAME
LEFT DC (MAIN), C4	CABIN PRESS AUTO CONT
LEFT DC (ESSENTIAL), L5	CABIN PRESS MAN

Subtask 21-00-00-800-001

- D. If you do this task to check the fuselage leak rate, plug the holes in the forward and aft drain masts with the rubber plugs as applicable.

Subtask 21-00-00-863-002

- E. Make sure that the door seal reservoir tank is pressurized. If necessary, pressurize the door seal reservoir tank (Refer to [TASK 12-10-52-863-801](#)).



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4. Procedure

**ON A/C 4001, 4006, 4008–4015, 4018–4037

Subtask 21–00–00–720–001

WARNING: MAKE SURE THAT THE CABIN DIFFERENTIAL PRESSURE IS NOT MORE THAN 5.8 ± 0.15 PSID (40 ± 1.0 KPAD). THE RELIEF VALVE CAN OPEN AT 5.8 ± 0.15 PSI, YOU CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO THE AIRCRAFT.

A. Do the pressurization of the aircraft as follows:

NOTE: The aircraft can be pressurized by either the left engine or the APU.

NOTE: The CABIN INDICATION MODULE (CIM) DIFF indicator will not show a differential pressure (shown as DIFF psi on the DIFF indicator) of more than 2.5 to 3 psiD (17.24 to 20.68 kPaD) on the ground. Use a calibrated pressure indicator (absolute pressure or gauge pressure) with a range from 0 to 25 psi (0 to 172.37 kPa) to determine the differential pressure values when the aircraft cabin is pressurized. Do not permit a pressure increase of more than 5.5 psiD (38.0 kPaD) above the internal ambient pressure prior to leakage test.

NOTE: The calibrated pressure indicator is referred to as the 'hand held pressure gauge' in this procedure. The pressure is measured in pounds per square inch (psi) or kilopascals (kPa) in this procedure.

If the hand held pressure gauge measures in inches of mercury (inHg), multiply the value by 0.49116 to get the psi value.

(1) Pressurize the aircraft with the left engine as follows:

- (a) Make sure all the doors are closed and locked.
- (b) On the CABIN ALTITUDE panel, set the AUTO/MAN/DUMP switch to the MAN position.
- (c) On the CABIN ALTITUDE panel, set the LDG ALT knob to get a LDG ALT of 0 ft.
- (d) Start the left engine (Refer to [TASK 71–00–00–868–801](#)) to supply a source of bleed air.
- (e) Set the left engine (No. 1) POWER lever to the IDLE or DISC position.
- (f) Set the right engine (No. 2) POWER lever to the RATING detent position at above 60 degrees angle.
- (g) Set the switches on the AIR CONDITIONING panel as follows:
 1. The BLEED 1 to the ON position.
 2. The BLEED flow control switch to the NORM position.
 3. The PACKS switches to the AUTO position.
- (h) On the CABIN ALTITUDE panel, move the MAN DIFF switch to the INCR position to pressurize the aircraft.



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- (i) On the CIM, make sure the RATE is not more than –300 FPM.
- (j) Closely monitor the pressure indications on the hand held pressure gauge. Make sure that the differential pressure indication on the hand held pressure gauge is not more than 5.5 psiD (38.0 kPaD).
- (k) If necessary, do an operational check of the fuselage leak rate as follows:

NOTE: The following steps 1 to 7 are necessary for a fuselage leak rate check only.

- 1. Pressurize the fuselage to a maximum pressure differential of 5.5 psiD (38.0 kPaD).
 - 2. Set the BLEED flow control switch to the MIN position.
 - 3. Hold the MAN DIFF switch to INCR for not more than 15 seconds. This will drive the outflow valve to full close position.
 - 4. Set the BLEED switch no.1 to the OFF position.
 - 5. Make sure that the time for the pressure to decrease from 5.0 to 3.0 psiD (34.5 to 20.7 kPaD) is not less than 194 seconds.
 - 6. If the time read is less than the specified limit then examine for leaks. If repair is necessary, complete steps (m) and (n) before you do the repair.
 - 7. If necessary, repeat the leak rate check.
- (l) Set the AUTO/MAN/DUMP switch to the AUTO position.
 - (m) When maintenance tasks are complete, depressurize the aircraft (Refer to [TASK 21-00-00-863-802](#)).
 - (n) Do not open aircraft doors or hatches until the internal pressure is dropped to 0 psiD (0 kPaD) (cabin ambient pressure) on the hand held pressure gauge.
- (2) Pressurize the aircraft with the APU as follows:
- (a) Make sure all the doors are closed and locked.
 - (b) On the CABIN ALTITUDE panel, set the AUTO/MAN/DUMP switch to the MAN position.
 - (c) On the CABIN ALTITUDE panel, set the LDG ALT knob to get a LDG ALT of 0 ft.
 - (d) Start the APU (Refer to [TASK 49-00-00-869-801](#)) to supply a source of bleed air.
 - (e) Set the right engine (No. 2) POWER lever to the RATING detent position at above 60 degrees angle.
 - (f) Set the PACKS switches to the AUTO position.
 - (g) On the AIR CONDITIONAL panel, make sure that:
 - 1. The BLEED 1 is set to OFF position.
 - 2. The BLEED 2 is set to OFF position.
 - 3. Push the APU BL AIR switch on the APU CONTROL panel. Make sure that the BL AIR OPEN light comes ON.



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- (h) On the CABIN ALTITUDE panel, move the MAN DIFF switch to the INCR position to pressurize the aircraft.
- (i) On the CIM, make sure the RATE is not more than -300 FPM.
- (j) Closely monitor the pressure indications on the hand held pressure gauge. Make sure that the differential pressure indication on the hand held pressure gauge is not more than 5.5 psiD (38.0 kPaD).
- (k) If necessary, do an operational check of the fuselage leak rate as follows:
NOTE: The following steps 1 to 6 are necessary for a fuselage leak rate check only.
 1. Pressurize the fuselage to a maximum pressure differential of 5.5 psiD (38.0 kPaD).
 2. Hold the MAN DIFF switch to INCR for not more than 15 seconds. This will drive the outflow valve to full close position.
 3. Push the APU BL AIR switch on the APU CONTROL panel. Make sure that the BL AIR OPEN light goes off.
 4. Make sure that the time for the pressure to decrease from 5.0 to 3.0 psiD (34.5 to 20.7 kPaD) is not less than 194 seconds.
 5. If the time read is less than the specified limit then examine for leaks. If repair is necessary, do steps (m) and (n) before you do the repair.
 6. If necessary, repeat the leak rate check.
- (l) Set the AUTO/MAN/DUMP switch to the AUTO position.
- (m) When maintenance tasks are complete depressurize the aircraft (Refer to [TASK 21-00-00-863-802](#)).
- (n) Do not open aircraft doors or hatches until the internal pressure is dropped to 0 psiD (0 kPaD) (cabin ambient pressure) on the hand held pressure gauge.

**ON A/C 4003-4004, 4016-4017, 4038-4140, 4142-4199, 4201-4999

Subtask 21-00-00-720-004

WARNING: MAKE SURE THAT THE CABIN DIFFERENTIAL PRESSURE IS NOT MORE THAN 5.8 ± 0.15 PSID (40 ± 1.0 KPAD). THE RELIEF VALVE CAN OPEN AT 5.8 ± 0.15 PSI, YOU CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO THE AIRCRAFT.

- B. Do the pressurization of the aircraft as follows:

NOTE: The aircraft can be pressurized by either the left engine or the APU.

- (1) Pressurize the aircraft with the left engine as follows:
 - (a) Make sure all the doors are closed and locked.
 - (b) On the CABIN ALTITUDE panel, set the AUTO/MAN/DUMP switch to the MAN position.



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- (c) On the CABIN ALTITUDE panel, set the LDG ALT knob to get a LDG ALT of 0 ft.
 - (d) Start the left engine (Refer to [TASK 71-00-00-868-801](#)) to supply a source of bleed air.
 - (e) Set the left engine (No. 1) POWER lever to the IDLE or DISC position.
 - (f) Set the right engine (No. 2) POWER lever to the RATING detent position at above 60 degrees angle.
 - (g) Set the switches on the AIR CONDITIONING panel as follows:
 - 1. The BLEED 1 to ON position.
 - 2. The BLEED FLOW CONTROL SWITCH to the NORM position.
 - 3. The PACKS switches to the AUTO position.
 - (h) On the CABIN ALTITUDE panel, move the MAN DIFF switch to the INCR position to pressurize the aircraft.
 - (i) On the CIM, make sure the RATE is not more than -300 FPM.
 - (j) Make sure that on the CIM indicator, the cabin differential pressure indication (shown on the indicator as DIFF psi) is not more than 5.5 DIFF psi.
 - (k) If necessary, do an operational check of the fuselage leak rate as follows:

NOTE: The following steps 1 to 7 are necessary for a fuselage leak rate check only.

 - 1. Pressurize the fuselage to a maximum pressure differential of 5.5 DIFF psi.
 - 2. Set the BLEED flow control switch to the MIN position.
 - 3. Hold the MAN DIFF switch to INCR for not more than 15 seconds. This will drive the outflow valve to full close position.
 - 4. Set the BLEED switch no.1 to the OFF position.
 - 5. Make sure that the time for the pressure to decrease from 5.0 DIFF psi to 3 DIFF psi is not less than 194 seconds.
 - 6. If the time read is less than the specified limit then examine for leaks. If repair is necessary, complete steps (m) and (n) before you do the repair.
 - 7. If necessary, repeat the leak rate check.
 - (l) Set the AUTO/MAN/DUMP switch to the AUTO position.
 - (m) When maintenance tasks are complete depressurize the aircraft (Refer to [TASK 21-00-00-863-802](#)).
 - (n) Do not open aircraft doors or hatches until the internal pressure indicated on the CIM has dropped to 0 DIFF psi (ambient cabin pressure).
- (2) Pressurize the aircraft with the APU as follows:
- (a) Make sure all the doors are closed and locked.



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- (b) On the CABIN ALTITUDE panel, set the AUTO/MAN/DUMP switch to the MAN position.
- (c) On the CABIN ALTITUDE panel, set the LDG ALT knob to get a LDG ALT of 0 ft.
- (d) Start the APU (Refer to [TASK 49-00-00-869-801](#)) to supply a source of bleed air.
- (e) Set the left engine (No. 1) POWER lever to the IDLE or DISC position.
- (f) Set the right engine (No. 2) POWER lever to the RATING detent position at above 60 degrees angle.
- (g) Set the PACKS switches to the AUTO position.
- (h) Set the switches on the AIR CONDITIONING panel as follows:
 - 1. The BLEED 1 to OFF position.
 - 2. The BLEED 2 to OFF position.
 - 3. Push the APU BL AIR switch on the APU CONTROL panel. Make sure that the BL AIR OPEN light comes ON.
- (i) On the CABIN ALTITUDE panel, move the MAN DIFF switch to the INCR position to pressurize the aircraft.
- (j) On the CIM, make sure the RATE is not more than -300 FPM.
- (k) Make sure that on the CIM indicator, the cabin differential pressure indication (shown on the indicator as DIFF psi) is not more than 5.5 DIFF psi.
- (l) If necessary, do an operational check of the fuselage leak rate as follows:

NOTE: The following steps 1 to 6 are necessary for a fuselage leak rate check only.

 - 1. Pressurize the fuselage to a maximum pressure differential of 5.5 DIFF psi.
 - 2. Hold the MAN DIFF switch to INCR for not more than 15 seconds. This will drive the outflow valve to full close position.
 - 3. Push the APU BL AIR switch on the APU CONTROL panel. Make sure that the BL AIR OPEN light goes off.
 - 4. Make sure that the time for the pressure to decrease from 5.0 DIFF psi to 3 DIFF psi is not less than 194 seconds.
 - 5. If the time read is less than the specified limit then examine for leaks. If repair is necessary, complete steps (n) and (o) before you do the repair.
 - 6. If necessary, repeat the leak rate check.
- (m) Set the AUTO/MAN/DUMP switch to the AUTO position.
- (n) When maintenance tasks are complete, depressurize the aircraft (Refer to [TASK 21-00-00-863-802](#)).
- (o) Do not open aircraft doors or hatches until the internal pressure indicated on the CIM has dropped to 0 DIFF psi (ambient cabin pressure).



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**ON A/C ALL

5. Close Out

Subtask 21-00-00-861-005

- A. Return all cockpit switches/selectors/controls to normal position.
- B. If installed, remove the rubber plugs from the forward and aft drain masts as applicable.
- C. De-energize the aircraft electrical system (Refer to [TASK 24-00-00-861-802](#)).

Subtask 21-00-00-941-003

- D. Remove all tools, equipment, and unwanted materials from the work area.



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AIRCRAFT MAINTENANCE MANUAL

TEMPORARY REVISION 21-125

FILING INSTRUCTIONS:

Insert this Temporary Revision after page 501 of AMM TASK 21-31-21-790-801.

Record the insertion of this Temporary Revision on the yellow RECORD OF TEMPORARY REVISION SHEET at the front of the Chapter.

Keep this Temporary Revision until normal revision service incorporates the information into the Aircraft Maintenance Manual.

DESCRIPTION:

This Temporary Revision revises AMM TASK 21-31-21-790-801 to update the latest illustration sheets to support the procedure.

PSM 1-84-2
EFFECTIVITY: See effectivity page(s) in TR# 21-125

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VALVE, FORWARD SAFETY – ADJUSTMENT/TEST

**ON A/C ALL

TASK 21-31-21-790-801**Purging and Leak Test of the Forward Safety Valve Pneumatic Lines**1. General

- A. The maintenance procedure that follows is for the purging and leak test of the forward safety valve pneumatic lines. The forward safety valve is installed in the nose compartment.

2. Job Set-Up Information

Subtask 21-31-21-943-002

A. Tools and Equipment

- | | | |
|-----|--|---|
| (1) | Commercially Available | Wrench, Torque 0 to 3000 lbf·in (0 to 338.95 N·m) |
| (2) | Commercially Available from Innovative Aircraft Solutions LLC. | Alternate Bleed Air Source (ABAS) Kit 63605-001 or Equivalent |
| (3) | Commercially Available | Plug AN806-4D |
| (4) | Commercially Available | Plug AN806-5D |
| (5) | Commercially Available | Plug AN815-5D |

Subtask 21-31-21-944-001

B. Consumable Materials

- | | | |
|-----|------------------------|-------------------------|
| (1) | Commercially Available | Dry air/shop air supply |
| (2) | Commercially Available | Leak Detector Solution |

Subtask 21-31-21-946-003

C. Reference Information



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REFERENCE	DESIGNATION
TASK 12-00-01-860-801	Standard Aircraft Configuration for Maintenance
TASK 21-31-00-710-805	Operational check of the Forward Safety Valve Opening with the Forward Safety Valve Selector
TASK 21-31-01-000-801	Removal of the Cabin Pressure Control Panel
TASK 21-31-01-400-801	Installation of the Cabin Pressure Control Panel
TASK 21-31-26-000-801	Removal of the Forward Safety Valve Selector
TASK 21-31-26-400-801	Installation of the Forward Safety Valve Selector
TASK 24-00-00-910-801	Electrical/Electronic Safety Precautions
TASK 31-61-01-000-801	Removal of the Display Units
TASK 31-61-01-400-801	Installation of the Display Units
TASK 31-61-01-710-801	Operational Test of the Multi-Function Display
TASK 31-61-01-710-803	Operational Test of the Engine Display

3. Job Set-Up

Subtask 21-31-21-860-003

- A. Make sure that the aircraft is in the standard configuration for maintenance (Refer to [TASK 12-00-01-860-801](#)).

Subtask 21-31-21-910-001

- B. Obey all the electrical/electronic safety precautions (Refer to [TASK 24-00-00-910-801](#)).

Subtask 21-31-21-865-003

- C. Open and tag the circuit breakers that follow:

CB PANEL AND CB NO.	NAME
LEFT DC (MAIN), C4	CABIN PRESS AUTO CONT
LEFT DC (ESSENTIAL), L5	CABIN PRESS MAN
LEFT (ESSENTIAL), B7	MFD1
RIGHT (ESSENTIAL), B11	ED



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Subtask 21-31-21-010-002

- D. Remove the EFIS display units (MFD1 and ED) (Refer to [TASK 31-61-01-000-801](#)).

4. Procedure

Subtask 21-31-21-790-001

[Refer to Figure 501.](#)

WARNING: MAKE SURE THAT YOU USE THE CORRECT SAFETY PROTECTION WHEN YOU PURGE THE LINES OF CONTAMINANTS. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO THE AIRCRAFT OR EQUIPMENT.

NOTE: Make sure that plugs are installed on all components after the hoses are removed to prevent contamination.

NOTE: Make sure that the sufficient safety protection is taken to prevent any personnel injury and safe collection of contaminants is done to avoid any damage to other parts of the aircraft or equipment during purging of the lines.

NOTE: For this test a regulated 15 psi (103.42 kPa) dry shop air supply may be used in lieu of the Alternate Bleed Air Source (ABAS) Kit.

- A. Do the steps that follow to purge the lines and clear the blockage:
- (1) Cabin Pressure Control Panel (2):
 - (a) Remove the cabin pressure control panel (2) (Refer to [TASK 21-31-01-000-801](#)).
 - (b) Put the plugs (AN806-5D and AN806-4D) at the hose ends.
 - (2) Forward Safety Valve (1):
 - (a) Remove the low pressure hose (4).
 - (b) Put the plug (AN806-4D) at the end.
 - (c) Remove the hoses from the pneumatic relay.
 - (d) Use a union (AN815-5D) to connect the ends.
 - (3) Suction Port (3):
 - (a) Remove the hose at the suction port (3).
 - (b) Put the plug (AN806-4D) at the open end.
 - (4) Forward Safety Valve Selector (6):
 - (a) Remove the forward safety valve selector (6) (Refer to [TASK 21-31-26-000-801](#)).
 - (b) Put the plug (AN806-4D) at the hose ends.
 - (5) Set up the ABAS Kit pressure at 15 psi (103.42 kPa).
 - (6) Remove the plug and connect the supply line at the test plug connector (5).



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- (7) Remove the plug at the suction port (3).
 - (8) Apply air pressure of 15 psi (103.42 kPa) and make sure that there is free flow of air out of the hose and no sign of contamination.
 - (9) Stop the air flow and put the plug.
 - (10) Remove the plug (AN806–5D) from the hose end at the cabin pressure control panel (2).
 - (11) Apply air pressure of 15 psi (103.42 kPa) and make sure that there is free flow of air out of the hose and no sign of contamination.
 - (12) Stop the air flow and put the plug.
 - (13) Remove the plug (AN806–4D) from the hose end at the cabin pressure control panel (2).
 - (14) Apply air pressure of 15 psi (103.42 kPa) and make sure that there is free flow of air out of the hose and no sign of contamination.
 - (15) Stop the air flow and put the plug.
 - (16) Remove the plug from the forward safety valve selector line coming from cabin pressure controller.
 - (17) Apply air pressure of 15 psi (103.42 kPa) and make sure that there is free flow of air out of the hose and no sign of contamination.
 - (18) Stop the air flow and install the forward safety valve selector (6) (Refer to [TASK 21–31–26–400–801](#)).
 - (19) Set the forward safety valve selector (6) to the OPEN position.
 - (20) Remove the plug (AN806–4D) from the low pressure hose (4) at the forward safety valve (1).
 - (21) Apply air pressure of 15 psi (103.42 kPa) and make sure that there is free flow of air out of the hose and no sign of contamination.
 - (22) Stop the air flow and put the plug.
 - (23) Remove the hose installed between the over pressure relief valve and the pressure port.
 - (24) Apply air pressure of 15 psi (103.42 kPa) and make sure that there is free flow of air out of the hose and no sign of contamination.
 - (25) Stop the air flow and install the hose.
- B. Do the leak test as follows:
- (1) Make sure that the ABAS Kit is connected to the test plug connector (5) and the pressure is set at 15 psi (103.42 kPa).
 - (2) Make sure that the forward safety valve selector (6) is in the OPEN position.
 - (3) Apply air pressure of 15 psi (103.42 kPa).
 - (4) Close the air supply valve and monitor the pressure for 15 minutes. The pressure must not decrease.



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- (5) If the pressure decreases, apply the leak detector fluid to all system joints to find the leak.

NOTE: Maintain air pressure to make sure that the leak detector solution does not go into the hoses.

- (6) If there is a air leakage, do as follows:

- (a) Clean off the leak detector solution.
- (b) Decrease air pressure to 0 psi (0 kPa).
- (c) Torque the connections which have threads to the maximum torque value (Refer to Table 1).

- NOTE:**
- 1. Use the minimum torque values at the time of the initial installation.
 - 2. Use only the correct dimension of wrenches to torque the nut. Incorrect tools can cause distortion of the nut.

Table 1: TORQUING COUPLING NUTS ON FLARED TUBES AND FITTINGS

TUBE SIZE		TORQUE VALUE					
		STRAIGHT THREADS ON ALUMINUM ALLOY TUBE OR FITTING (Notes 1 & 2)		STRAIGHT THREADS ON STEEL ALLOY, CRES OR TITANIUM TUBE OR FITTINGS (Note 1)		TAPERED PIPE THREADS (except oxygen & pitot systems)	
in.	mm	lbf-in	N·m	lbf-in	N·m	lbf-in	N·m
1/8	3.17	20 – 30	2.25 – 3.38	75 – 85	8.47 – 9.60	40 – 300	4.51 – 33.89
3/16	4.76	25 – 35 (15 – 35)	2.82 – 3.95 (1.69 – 3.95)	95 – 105	10.73 – 11.86		
1/4	6.35	50 – 65 (20 – 65)	5.64 – 7.34 (2.25 – 7.34)	135 – 150	15.25 – 16.94	60 – 600	6.77 – 67.79
5/16	7.93	70 – 90 (30 – 90)	7.90 – 10.16 (3.38 – 10.16)	170 – 200	19.20 – 22.59		
3/8	9.52	110 – 130 (35 – 130)	12.42 – 14.68 (3.95 – 14.68)	270 – 300	30.50 – 33.89	75 – 700	8.47 – 79.08
1/2	12.7	230 – 260	25.98 – 29.37	450 – 500	50.84 – 56.49	100 – 900	11.29 – 101.68
5/8	5.87	330 – 360	37.28 – 40.67	650 – 700	73.44 – 79.08		
NOTE: 1. Use the lower torque value if the hose fitting material and the tube material are different. 2. Torque values in brackets apply to 5052-O aluminum tubes with a thickness of 0.022 in. (0.558 mm).							



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Table 1: TORQUING COUPLING NUTS ON FLARED TUBES AND FITTINGS

TUBE SIZE		TORQUE VALUE					
		STRAIGHT THREADS ON ALUMINUM ALLOY TUBE OR FITTING (Notes 1 & 2)		STRAIGHT THREADS ON STEEL ALLOY, CRES OR TITANIUM TUBE OR FITTINGS (Note 1)		TAPERED PIPE THREADS (except oxygen & pitot systems)	
in.	mm	lbf·in	N·m	lbf·in	N·m	lbf·in	N·m
3/4	19.05	460 – 500	51.97 – 56.49	900 – 1000	101.68 – 112.98	300 – 1600	33.89 – 180.77
1	25.4	500 – 700	56.49 – 79.08	1200 – 1400	135.58 – 158.17	400 – 2200	45.19 – 248.56
1 – 1/4	25.4 – 6.35	800 – 900	90.38 – 101.68	1520 – 1680	171.73 – 189.81	500 – 2500	56.49 – 282.46
1 – 1/2	25.4 – 12.7	800 – 900	90.38 – 101.68	1900 – 2100	214.67 – 237.26		
2	50.8	1800 – 2000	203.37 – 225.96	2660 – 2940	300.53 – 332.17		
NOTE: 1. Use the lower torque value if the hose fitting material and the tube material are different. 2. Torque values in brackets apply to 5052-O aluminum tubes with a thickness of 0.022 in. (0.558 mm).							

(7) If the leak continues in the fitting, do as follows:

- (a) Decrease the torque in the fittings.
- (b) Torque the fittings to the values specified in table (2).

Table 2: LEAK-PROOF TORQUE VALUES FOR FLARED COUPLING NUTS

FITTING SIZE		TORQUE VALUE			
		ALUMINUM FITTING		CRES FITTING	
in.	mm	lbf·in	N·m	lbf·in	N·m
1/4	6.35	157 – 173	17.73 – 19.54	200 – 220	22.59 – 24.85
3/8	9.52	242 – 268	27.34 – 30.27	380 – 420	42.93 – 47.45
1/2	12.7	400 – 440	45.19 – 49.71	713 – 787	80.55 – 88.91
5/8	15.87	513 – 567	57.96 – 64.06	998 – 1102	112.75 – 124.5
3/4	19.05	641 – 708	72.42 – 79.99	1283 – 1417	144.95 – 160.09
1	25.4	1069 – 1181	120.78 – 133.43	1710 – 1890	193.2 – 213.54

(c) If the leak continues, replace the fitting or the line.



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- (d) Discard over-torqued fittings. Do not loosen and tighten an over-torqued fitting.
- (e) Make sure that flexible hoses and rigid lines are not twisted. It can show an incorrect torque value indication and subsequent leakage.
- (8) If there is no leakage, do as follows:
 - (a) Decrease air pressure to 0 psi (0 kPa).
 - (b) Disconnect and remove the ABAS Kit from the shop air source and the test plug connector (5).
 - (c) Put the cap to the test plug connector (5).
 - (d) Set the forward safety valve selector (6) to the NORMAL position.
- C. Restore the system configuration as follows:
 - (1) Cabin Pressure Control Panel (2):
 - (a) Remove the plugs from hoses ends.
 - (b) Install the cabin pressure control panel (2) (Refer to [TASK 21-31-01-400-801](#)).
 - (2) Forward Safety Valve (1):
 - (a) Remove the union fitting and connect the lines at the pneumatic relay.
 - (b) Remove the plug and connect the line from the forward safety valve selector (6).
 - (c) Connect the hose from the Pressure Port to the Over Pressure Relief line.
 - (3) Suction port (3):
 - (a) Remove the plug.
 - (b) Reconnect the line at the suction port (3).
 - (c) Set the forward safety valve selector (6) to the NORMAL position.

5. Close-Out

Subtask 21-31-21-410-004

- A. Install the EFIS display unit (MFD1 and ED) (Refer to [TASK 31-61-01-400-801](#)).

Subtask 21-31-21-865-004

- B. Remove the tags and close the circuit breakers that follow:

CB PANEL AND CB NO.	NAME
LEFT DC (MAIN), C4	CABIN PRESS AUTO CONT
LEFT DC (ESSENTIAL), L5	CABIN PRESS MAN
LEFT (ESSENTIAL), B7	MFD1



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RIGHT (ESSENTIAL), B11	ED
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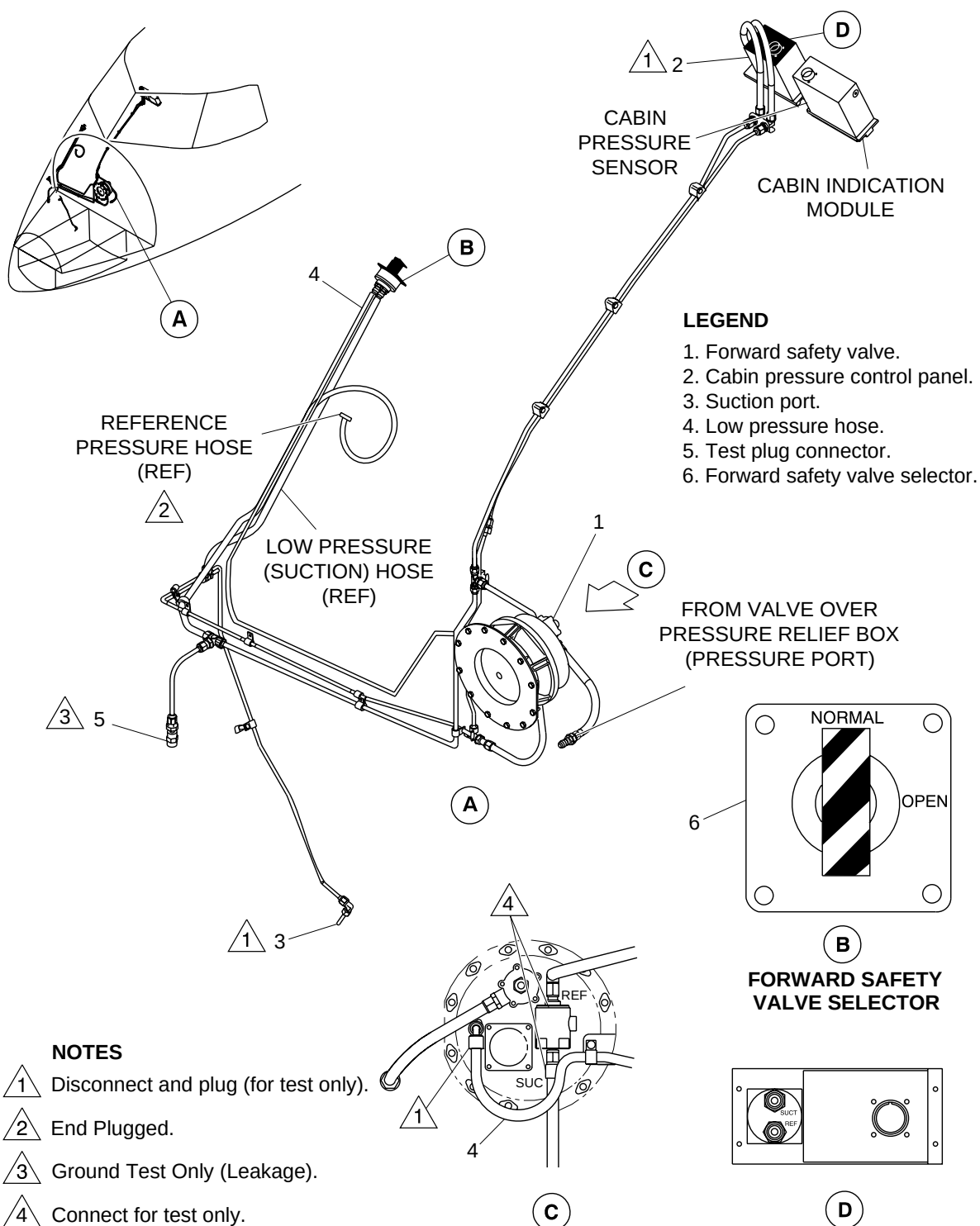
Subtask 21-31-21-710-003

- C. Do an operational check of the EFIS display unit (MFD1) (Refer to [TASK 31-61-01-710-801](#)).
- D. Do an operational check of the EFIS display unit (ED) (Refer to [TASK 31-61-01-710-803](#)).
- E. Do an operational check of the Forward Safety Valve Opening with the Forward Safety Valve Selector (Refer to [TASK 21-31-00-710-805](#)).

Subtask 21-31-21-941-005

- F. Remove all tools, equipment, and unwanted materials from the work area.

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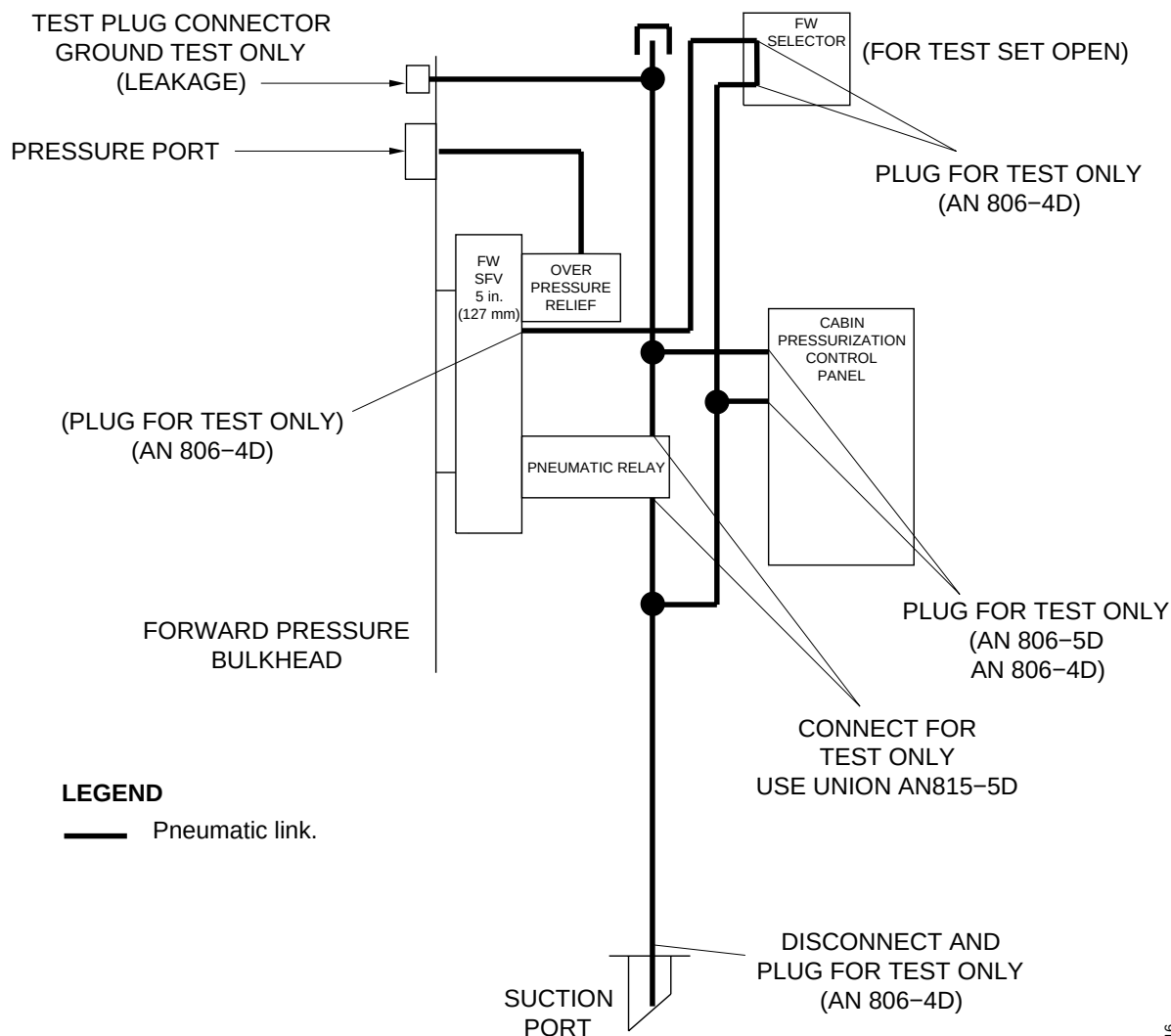


Leak Test of the Forward Safety Valve Pneumatic Lines

Figure 501 (Sheet 1 of 2)



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Leak Test of the Forward Safety Valve Pneumatic Lines
Figure 501 (Sheet 2 of 2)



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